

Towards Improving Accessibility of Large Documents

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In the context of Natural Language Processing (NLP), accessibility refers to computational methods that transform complex, technical, and information-dense documents into forms that are easier to understand, navigate, and use by both humans and machines, while preserving their original meaning. This involves restructuring jargon-heavy and unstructured text, enabling efficient retrieval of relevant information, clarifying relationships within documents, and supporting automated reasoning and decision-making. Formal documents such as legal, medical, and scientific texts present unique challenges due to their domain-specific terminology, dense semantic content, and long-range dependencies, making them difficult for non-experts to comprehend and for general-purpose NLP models to process effectively. Key information is often distributed across sections and expressed using specialized language, making manual review time-consuming and cognitively demanding even for experts. Improving accessibility addresses these challenges by enabling faster information retrieval, clearer interpretation, and more efficient analysis, while also allowing automated systems to extract knowledge and generate meaningful insights. Ultimately, accessibility in NLP serves as a bridge between complex domain-specific knowledge and usable, human-centered understanding, making critical information both comprehensible and actionable.

Although many documents appear unstructured on the surface, the formality of language and the constraints of a specific domain often introduce an implicit structure that can be systematically analyzed. Building on this observation, we first investigated the internal organization of contracts to better support their accessibility and analysis. The key findings indicate that contracts generally contain three key types of textual components: (a) Templatized sentences, which maintain a consistent structure while inserting contextual values specific to the agreement; (b) Boilerplate clauses, which are standard across contracts of a particular type and often constitute the majority of the document; and (c) Rare clauses, which appear infrequently, deviate from common patterns, and often signal content of particular significance or risk. Based on these insights the study in SConE [2] highlighted key sections of interest within the contract. This approach facilitated the visualization of significant components and helped surface novel or non-standard clauses that deviate from expected norms. The study not only provided a deeper understanding of the structural organization of contracts but also laid the groundwork for identifying subsequent accessibility-focused tasks and informed a more mature, principled approach to developing tools that support clearer, and more effective contract interpretation.

Entity extraction is another key task, traditionally addressed through named entity recognition (NER), which identifies entities such as names, dates, and organizations. However, legal entities often require fine-grained, context-specific interpretation, making manual extraction costly and inefficient. To address this, we reframe NER as an instruction-based generation task using large language models (LLMs). By incorporating task-specific instructions and constrained outputs, our approach [1] improves extraction accuracy while reducing hallucinations, enhancing both reliability and adaptability.

Rhetorical role labeling further improves document understanding by identifying the functional role of each sentence (e.g., facts, arguments, decisions), enabling structured interpretation and supporting downstream tasks such as summarization and classification. Given the complexity of legal texts, we automate this task using domain-adapted models. Our work [3], developed as part of SemEval-2023 Task 6 [4], explores domain-specific embeddings and neural classifiers, demonstrating the effectiveness of fine-tuned LegalBERT representations.

Text simplification forms a central contribution of this thesis toward improving the accessibility of legal documents, which are often dense and written using specialized vocabulary and complex sentence structures that limit comprehension for non-expert audiences. Existing methods typically produce a single simplified version, assuming a uniform audience, whereas in practice users require different levels of simplification based on their expertise. To address this, we introduce Graded Simplification for Legal Data (GSLD), a framework that generates paragraph-level simplifications across three levels: Skilled, Intermediate, and Basic, tailored to diverse users while preserving legal meaning and validity using large language models. To incorporate broader document context efficiently, we propose Token-efficient Compression (TeC), which produces compact contextual representations that retain essential information without high computational cost. We also develop a multi-criteria evaluation framework measuring readability, lexical simplicity, semantic preservation, and entailment, and introduce the SimpLegal dataset, a preference-based resource that enables preference optimization methods such as Direct Preference Optimization to improve simplification quality. Collectively, these contributions demonstrate that graded, context-aware simplification can significantly improve legal accessibility and provide an efficient, scalable solution for generating legally faithful and audience-appropriate simplified text.

Beyond simplification and entity extraction, this thesis will investigate structured summarization of contracts into tabular formats that capture actionable information such as Permissions, Obligations, and Prohibitions, which are typically dispersed across complex clauses. By automatically generating these structured summaries, the proposed approach will enable users to quickly identify key responsibilities and constraints without reviewing entire documents. This task will be implemented using large language models, retrieval-augmented generation (RAG), and multi-agent frameworks, where retrieval will ensure factual grounding and specialized agents will improve reliability and interpretability. Overall, this structured representation will transform contracts from unstruc-

tured text into actionable knowledge, improving efficiency in contract review and compliance analysis.

In addition to textual and tabular representations, this thesis will explore visual representations of technical documents through the automatic generation of mind maps. Mind maps provide hierarchical, visual summaries that capture relationships between concepts, clauses, and entities. Converting complex legal, scientific, or medical documents into mind maps enables users to quickly understand document structure, identify key components, and navigate relationships between concepts. This work proposes methods for transforming unstructured text into structured visual representations by extracting hierarchical relationships, key concepts, and dependencies using large language models and structured reasoning frameworks. These visual summaries enhance accessibility by reducing cognitive load and enabling intuitive exploration of complex information. Furthermore, such representations support tasks such as legal review, scientific literature analysis, and educational applications by presenting information in a format that is easier to interpret and analyze.

While this work focuses on legal documents, the challenges addressed are common across domains such as medicine, finance, and scientific research. The proposed methods—including graded simplification, instruction-based extraction, contextual compression, structured summarization, and visual representations—offer a general framework for improving accessibility. Overall, this research aims to bridge the gap between complex domain-specific knowledge and usable, human-centered understanding, making information more interpretable and actionable.

References

1. ADIBHATLA, H., BASWANI, P., AND SHRIVASTAVA, M. Fine-grained contract ner using instruction based mode. In *Proceedings of the 37th Pacific Asia Conference on Language, Information and Computation* (2023), pp. 889–902.
2. ADIBHATLA, H., AND SHRIVASTAVA, M. SConE:contextual relevance based Significant CompoNent Extraction from contracts. In *Proceedings of the 19th International Conference on Natural Language Processing (ICON)* (New Delhi, India, Dec. 2022), M. S. Akhtar and T. Chakraborty, Eds., Association for Computational Linguistics, pp. 161–171.
3. BASWANI, P., ADIBHATLA, H. S., AND SHRIVASTAVA, M. Ltrc at semeval-2023 task 6: experiments with ensemble embeddings. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)* (2023), pp. 841–846.
4. MODI, A., KALAMKAR, P., KARN, S., TIWARI, A., JOSHI, A., TANIKELLA, S. K., GUHA, S. K., MALHAN, S., AND RAGHAVAN, V. Semeval 2023 task 6: Legaleval-understanding legal texts. *arXiv preprint arXiv:2304.09548* (2023).